

Version Control Using Git and GitHub

Project Description

This project demonstrates the implementation of Version Control System using Git and GitHub. The project covers essential Git operations such as repository creation, branch management, committing changes, merging branches, rebasing, pull requests, and remote repository synchronization. The main objective of this project is to understand collaborative development workflow and source code management using Git.

Features

- Git repository initialization
- Local and remote repository connection
- Branch creation and branch switching
- Commit creation and commit history tracking
- Merge and rebase operations
- Pull request creation and branch comparison
- GitHub repository synchronization
- Conflict handling and project version management

EXPLORER

VERSION-CONTORL-GIT PROJECT

templates

index.html

app.py

data.json

Welcome

app.py

data.json

templates

index.html

app.py > ...

4

5 app = Flask(__name__)

6

7 client = MongoClient("mongodb+srv://Admin:Admin123@cluster1.jumillq.mongodb.net/?appName=Cluster1")

8

9 db = client["todoDB"]

10

11 collection = db["items"]

12

13 @app.route('/')

14 def home():

15 | return render_template('index.html')

16 |

17 @app.route('/api')

18 def api():

19 |

20 | with open('data.json') as file:

21 | | data = json.load(file)

22 |

23 | return jsonify(data)

24

25 @app.route('/submittodoitem', methods=['POST'])

26 def submit():

27 |

28 | itemName = request.form.get('itemName')

29 |

30 | itemDescription = request.form.get('itemDescription')

31

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

PS D:\Version-contorl-git project> git push origin main --force

Compressing objects: 100% (4/4), done.

Writing objects: 100% (4/4), 839 bytes | 279.00 KiB/s, done.

Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To https://github.com/shaktibehera149-boop/Version-control-git-project.git

+ 2a12753...0b1ac3e main -> main (forced update)

● PS D:\Version-contorl-git project> git checkout master_1

Switched to branch 'master_1'

● PS D:\Version-contorl-git project> git rebase main

Successfully rebased and updated refs/heads/master_1.

● PS D:\Version-contorl-git project> git push origin master_1 --force

Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)

To https://github.com/shaktibehera149-boop/Version-control-git-project.git

9b33c68..0b1ac3e master_1 -> master_1

● PS D:\Version-contorl-git project> git checkout main

Switched to branch 'main'

Your branch is up to date with 'origin/main'.

● PS D:\Version-contorl-git project> git push origin main

Everything up-to-date

○ PS D:\Version-contorl-git project>

> OUTLINE

> TIMELINE

< main

0 0 0

Shakti Kumar Behera (1 hour ago)

Ln 16, Col 1

Spaces: 4

UTF-8

CRLF

{ } Python

3.11.9 (Microsoft Store)

Python 3.11

Go Live

Prettier

Steps Performed in the Project

Step 1: Repository Creation

Created a new repository on GitHub for project version control.

Step 2: Local Project Setup

Opened the project folder in Visual Studio Code and initialized Git using:

Bash

```
git init
```

Step 3: Remote Repository Connection

Connected the local project with GitHub repository:

Bash

```
git remote add origin REPOSITORY_LINK
```

Step 4: Initial Commit

Added project files and created the first commit:

Bash

```
git add .
```

```
git commit -m "Initial commit"
```

ToDo Form

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS D:\Version-contorl-git project> python app.py
```

```
* Running on http://127.0.0.1:5000
```

```
Press CTRL+C to quit
```

```
* Restarting with stat
```

```
* Debugger is active!
```

```
* Debugger PIN: 870-582-088
```

```
❖ PS D:\Version-contorl-git project> ^C
```

```
○ PS D:\Version-contorl-git project> python app.py
```

```
* Running on http://127.0.0.1:5000
```

Step 6: Branch Creation

Created new branches for feature updates:

Bash

```
git checkout -b master_1
```

```
git checkout -b master_2
```

Step 7: Branch Management

Switched between branches and updated project files:

Bash

```
git checkout main
```

```
git checkout master_1
```

Step 8: Merge and Rebase

Performed merge and rebase operations to synchronize branches:

Bash

```
git merge master_1
```

```
git rebase main
```


shakti

shakti

Submit

```
PS D:\Version-contorl-git project> python app.py
File "C:\Users\shakt\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8
_validate_uri
    user, passwd = parse_userinfo(userinfo)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
File "C:\Users\shakt\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.11_qbz5n2kfra8
parse_userinfo
    raise InvalidURI(
pymongo.errors.InvalidURI: Username and password must be escaped according to RFC 3986, use ur
PS D:\Version-contorl-git project> python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a product
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 870-582-088
127.0.0.1 - - [07/May/2026 23:00:14] "GET / HTTP/1.1" 200 -
```

```

PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\Version-control-git project> git push origin main --force
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 839 bytes | 279.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shaktibehera149-boop/Version-control-git-project.git
+ 2a12753...0b1ac3e main -> main (forced update)
● PS D:\Version-control-git project> git checkout master_1
Switched to branch 'master_1'
● PS D:\Version-control-git project> git rebase main
Successfully rebased and updated refs/heads/master_1.
● PS D:\Version-control-git project> git push origin master_1 --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shaktibehera149-boop/Version-control-git-project.git
9b33c68..0b1ac3e master_1 -> master_1
● PS D:\Version-control-git project> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
● PS D:\Version-control-git project> git push origin main
Everything up-to-date
○ PS D:\Version-control-git project> █

```

> **OUTLINE**
 > **TIMELINE**
 main

✉ Shakti Kumar Behera

Step 9: Pull Request and GitHub Sync

Created pull requests and pushed updated changes to GitHub:

Bash

```
git push origin main
```

Outcome

The project successfully demonstrated Git version control workflow including branch handling, repository synchronization, merge operations, rebasing, and GitHub collaboration features.

THANK YOU!

Feel free to approach us if you have any questions.